

be observed. Hence line spectra is evidence for existence of energy levels with atoms

28 Ans: D

Uncertainty in the position of the photon:

$$\Delta x = c \times t = (3.00 \times 10^8)(3.0 \times 10^9) = 0.90 \text{ m}$$

$$\Delta x \Delta p \geq h$$

$$\Delta p = \frac{h}{0.90} = 7.4 \times 10^{-34} \text{ kg m s}^{-1}$$

